Agent-based Modeling and Simulation

3(2, 1)

Week 6

Dr. Muhammad Adeel

Purposes of the Declared Variables …

- **metastatic?:** Only true or false can be stored in this variable. Actually there are two types of Stem cells in this model:
  1. Stem cell 0
  2. Stem cell 1

  Further both types of Stem cells have progeny (children) cells. Now since both are stem cells therefore we need to keep the track of their progeny cells to distinct them. If we store “true” in metastatic variable it means cell* is progeny of “Stem cell 1” and if we store “false”, then it means cell is progeny of “Stem cell 0”.
- **cell-count:** To store, show and control the number of created cells during execution of the model.

*for which metastatic variable is being used

```
turtles-own
[
  stem?     ;; true for stem cells, false for transitory cells
  age  ;; age of cell. changes color with age
  metastatic?  ;; false for progeny of stem cell 0, true for progeny of stem cell 1
]

globals
[
  cell-count
]

to setup
  clear-all
  set-default-shape turtles "circle 3"
  ask patches
    [ set pcolor gray ]
  set-stem
  evaluate-params
  reset-ticks
end

to set-stem   ;;create two stem cells
  crt 2
  [
    set size 2   ; easier to see
    setxy (min-pxcor / 2) 0
    set stem? true
    set metastatic? false
    set color blue
    set age 0
  ]
  ask turtle 1
  [
    set metastatic? true
    set heading 90              ;; stem cell 1 will move away
  ]
  set cell-count 2
end
```

setup and set-stem are names of two functions

We use to and end keywords for defining body (boundaries) of a function

```
to go
  ask turtles
  [
    ifelse leave-trail?
      [ pd ]
      [ pu ]
    if (who = 1) and (xcor < 25)
    [ fd 1 ]  ;stem cell movement
    set age age + 1
    move-transitional-cells
    mitosis
    death
  ]
  tick
  evaluate-params
end
;;transitional cells move and hatch more. Turtle proc.
to move-transitional-cells
  if (not stem?)
  [
    set color ( red + 0.25 * age )
    fd 1
    if (age < 6)
    [
      hatch 1
      [  ;amplification
        rt random-float 360
        fd 1
      ]
    ]
  ]
end
to mitosis ;; turtle proc. - stem cells only
  if stem?
  [
    hatch 1
    [
      fd 1
      set color red
      set stem? false
      ifelse (who = 1)
        [ set age 16 ]
        [ set age 0 ]
    ]
  ]
end
```

go, move-transitional-cells and mitosis are names of three functions

```
to death   ;; turtle proc.
  if (not stem?) and (not metastatic?) and (age > 20)
    [ die ]
  if (not stem?) and metastatic? and (age > 4)
    [ die ]
end

to evaluate-params
  set cell-count count turtles  ;cell count
  if (cell-count <= 0)
    [ stop ]
end

to kill-original-stem-cell
  ask turtle 0
    [ die ]
end

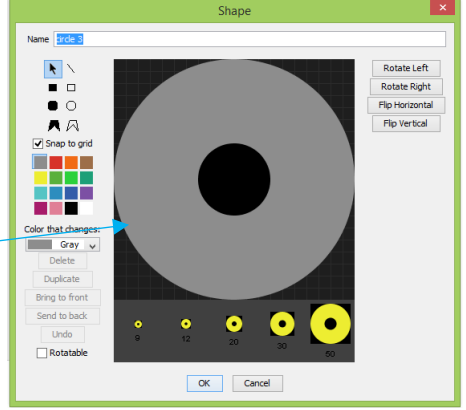
to kill-moving-stem-cell
  ask turtle 1
    [ die ]
end

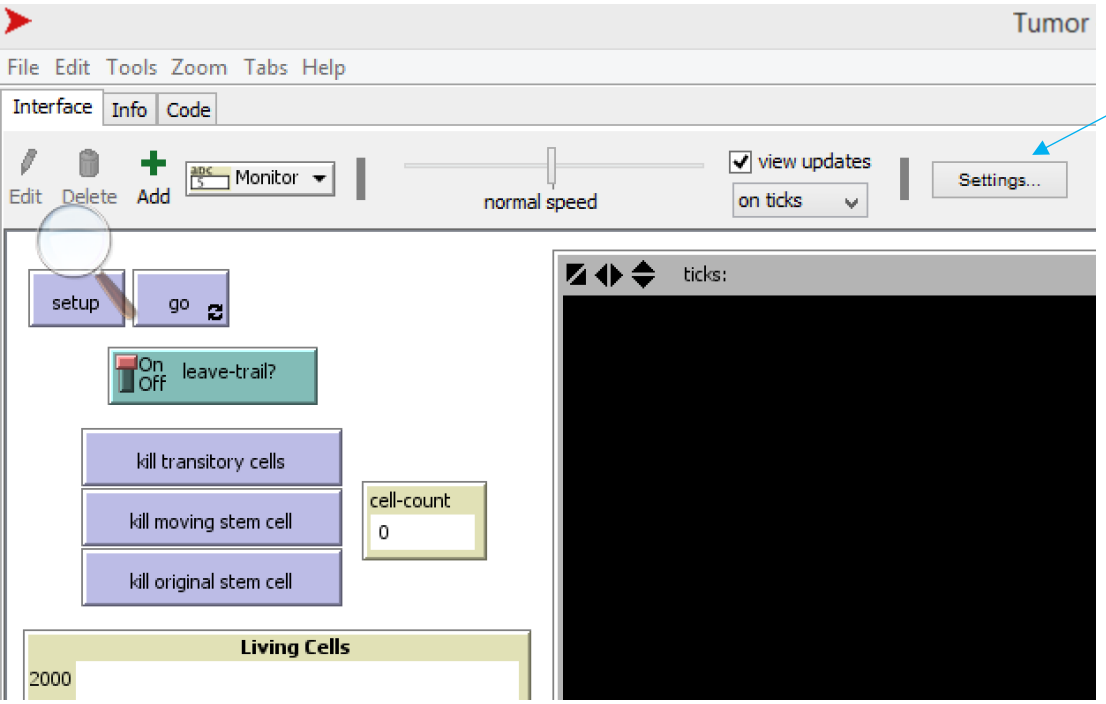
to kill-transitory-cells
  ask turtles with [ age < 10 and not stem? ]
    [ die ]
end
```

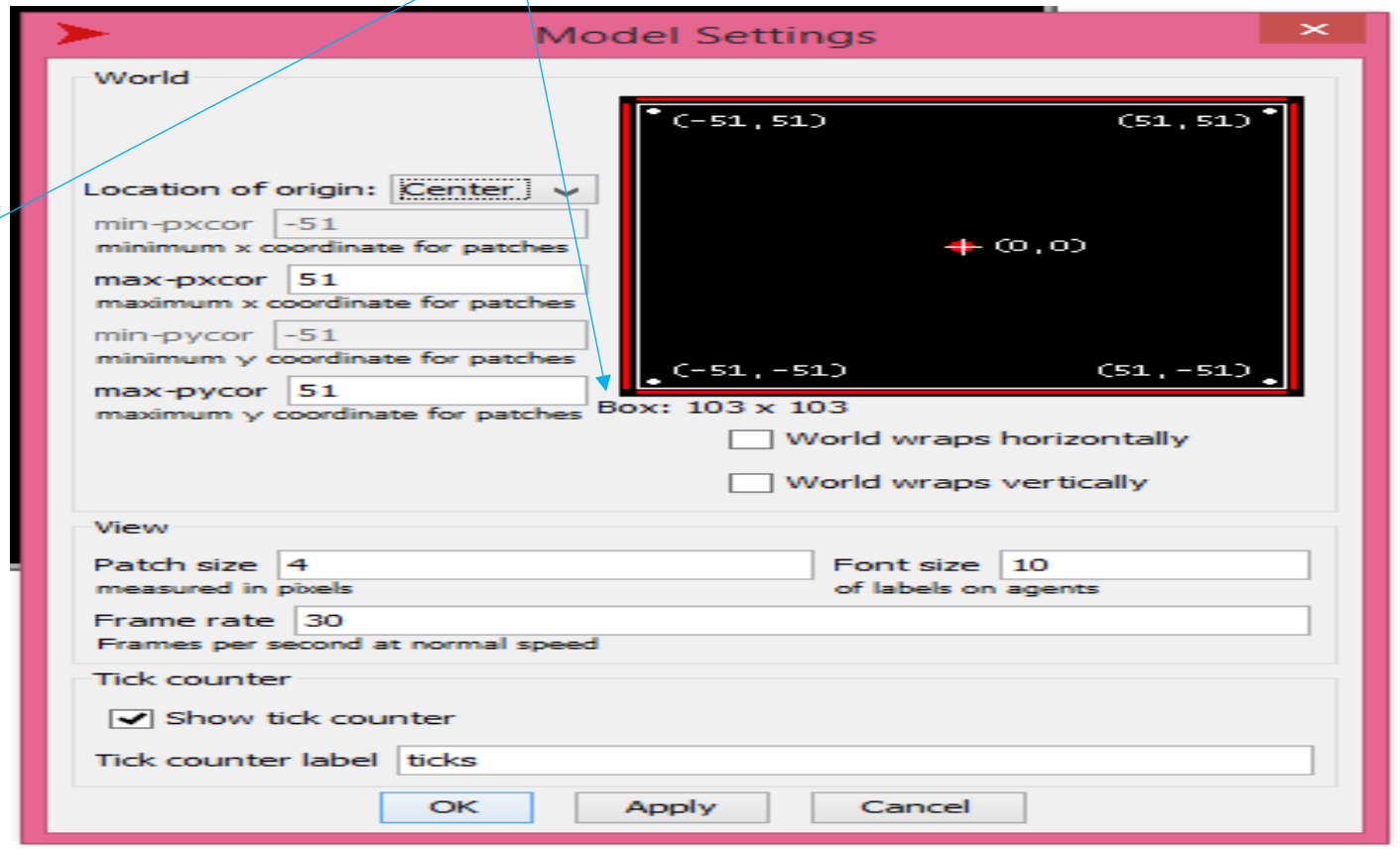
death, evaluate-params, kill-original-stem-cell, kill-moving-stem-cell and kill-transitory-cells are names of five functions

Agents used in this model

- There are two agents used in this model:
  1. Turtle (shape is “circle 3”)
  2. Patches (back-ground has been equally divided into 10609 grids called patches, we can see the patches settings by clicking on setting button)

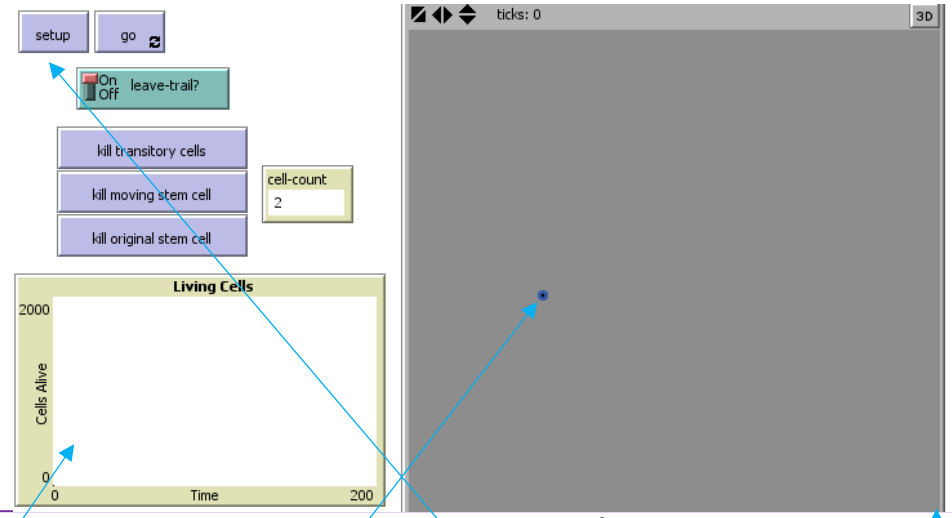






Understanding “setup” function

```
1.....to setup
2.....clear-all
3.....set-default-shape turtles "circle 3"
4.....ask patches
5.....  [ set pcolor gray ]
6.....set-stem
7.....evaluate-params
8.....reset-ticks
9.....end
```



1: This line gives a name “setup” to function (which is called when user clicks the setup button).

2: It clears all things of the world where we see output of our agents

3: It sets turtles shape “circle 3”, whereas by-default the turtle shape is:



4: It alerts patches to execute commands with-in next brackets.

5: It gives the gray color to all patches that results the world background gray.

6: It calls “set-stem” function (which basically displays two blue-colored turtles in the world, see its explanation on next slide).

7: It calls “evaluate-params” function (which basically counts and save created number of cells and stops program execution if the number of current cells are less than or equal to zero)

8: It resets the tick counter to zero, sets up the plot, then updates the plots (so that the initial state of the world is plotted)

9: It ends the function body.

Understanding “set-stem” function

```
1  to set-stem   ;;create two stem cells
2    crt 2
3    [
4      set size 2    ; easier to see
5      setxy (min-pxcor / 2) 0
6      set stem? true
7      set metastatic? false
8      set color blue
9      set age 0
10   ]
11   ask turtle 1
12   [
13     set metastatic? true
14     set heading 90
15   ]
16   set cell-count 2
17 end
```

2: This line creates two turtles

3: It alerts NetLogo to be ready to set the properties of the created two turtles.

Understanding "set-stem" function …

```
1  to set-stem   ;;create two stem cells
2    crt 2
3    [
4      set size 2    ; easier to see
5      setxy (min-pxcor / 2) 0
6      set stem? true
7      set metastatic? false
8      set color blue
9      set age 0
10   ]
11   ask turtle 1
12   [
13     set metastatic? true
14     set heading 90
15   ]
16   set cell-count 2
17 end
```

4: It increases size of the created turtles to the size of two patches (by-default turtle size is equal to one patch)

5: min-pxcor is used to get minimum x coordinate of patch of the world of. Its value for this model is -51, so divided by 2 gives -25.5. setxy (with two values x and y) is used to set the location of the turtle , so Line 5 sets the location of the created two turtles at (-25.5, 0).

Understanding "set-stem" function …

```
1  to set-stem    ;;create two stem cells
2    crt 2
3    [
4      set size 2    ; easier to see
5      setxy (min-pxcor / 2) 0
6      set stem? true
7      set metastatic? false
8      set color blue
9      set age 0
10   ]
11   ask turtle 1
12   [
13     set metastatic? true
14     set heading 90
15   ]
16   set cell-count 2
17 end
```

- 6: It saves the value "true" in the stem variable of the two created turtles that keeps the track of the cells i.e., which cell is stem cell and which is not stem cell, in this case true has been saved means the two created turtles are stem cell (and if we pass false then it means these are not stem cells but they are other type of cells(transitory cells, see Slide 19 for details))

Understanding “set-stem” function …

```
1  to set-stem   ;;create two stem cells
2    crt 2
3    [
4      set size 2    ; easier to see
5      setxy (min-pxcor / 2) 0
6      set stem? true
7      set metastatic? false
8      set color blue
9      set age 0
10   ]
11   ask turtle 1
12   [
13     set metastatic? true
14     set heading 90
15   ]
16   set cell-count 2
17 end
```

7: It saves the value “false” in the metastatic variable that means the two created turtles are progeny (children) of stem cell 0. (see Slide 20 for details)

8: It gives blue color to the two created cells (turtles)

9: It stores zero in age variable (see Slide 19 for details)

Understanding “set-stem” function …

```
1  to set-stem   ;;create two stem cells
2    crt 2
3    [
4      set size 2    ; easier to see
5      setxy (min-pxcor / 2) 0
6      set stem? true
7      set metastatic? false
8      set color blue
9      set age 0
10   ]
11   ask turtle 1
12   [
13     set metastatic? true
14     set heading 90
15   ]
16   set cell-count 2
17 end
```

10: It tells the NetLogo that settings of properties of the two created cells have been completed

11: It alerts Turtle no. 1 out of the two created turtles to be ready to set its properties with-in the coming brackets.

13: It saves the value “true” in the metastatic variable that means the Turtle No. 1 is progeny (children) of stem cell 1.

Understanding "set-stem" function …

```
1  to set-stem   ;;create two stem cells
2    crt 2
3    [
4      set size 2    ; easier to see
5      setxy (min-pxcor / 2) 0
6      set stem? true
7      set metastatic? false
8      set color blue
9      set age 0
10   ]
11   ask turtle 1
12   [
13     set metastatic? true
14     set heading 90
15   ]
16   set cell-count 2
17 end
```

14: It turns the turtle 1 to east from its current direction (heading is a built-in turtle variable. It indicates the direction the turtle is facing. This is a number greater than or equal to 0 and less than 360. 0 is north, 90 is east, and so on. You can set this variable to make a turtle turn.)

16: It stores 2 in cell-count variable (see Slide 20 for details).

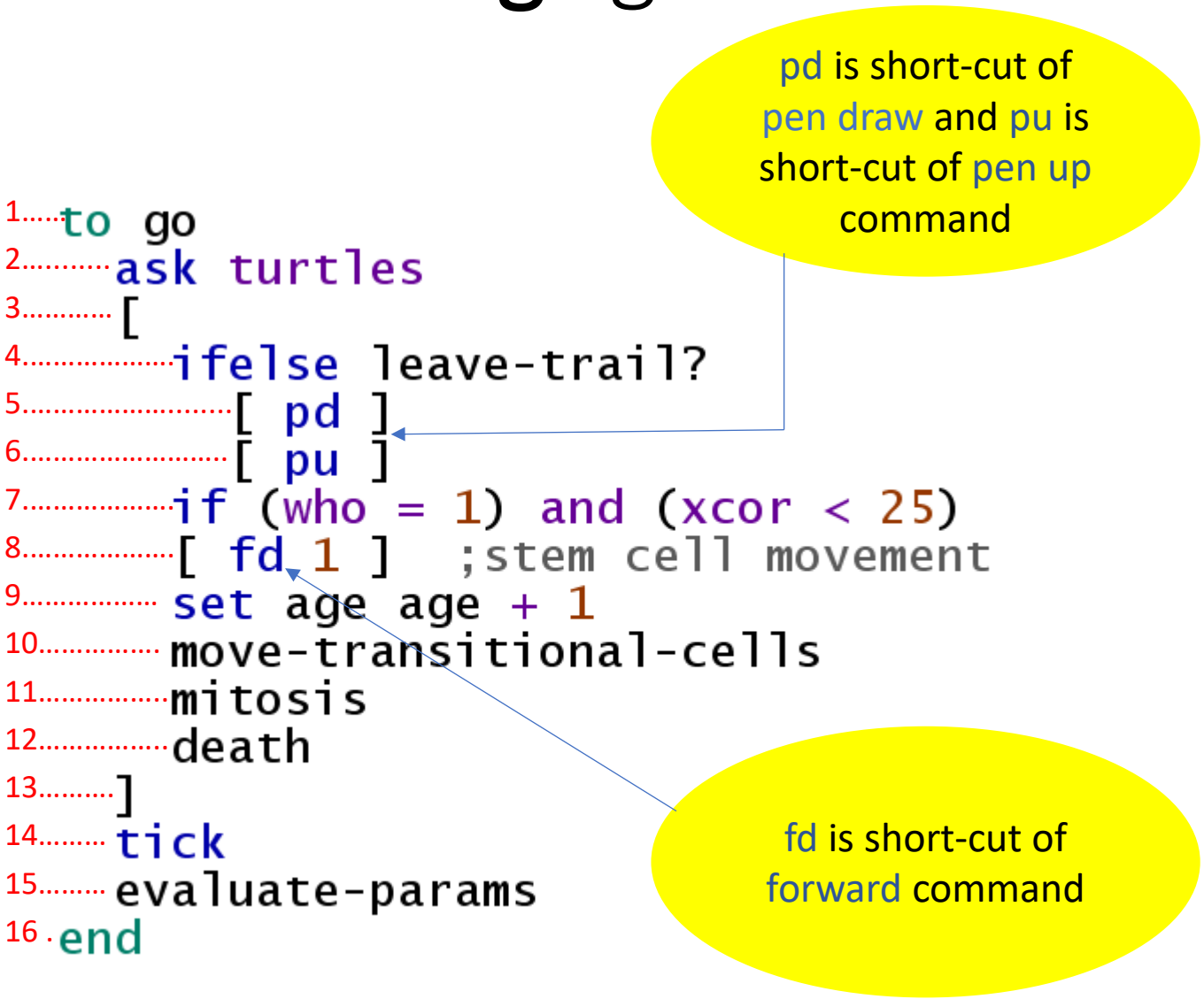
Understanding “evaluate-prams” function

```
1……to evaluate-params
2…… set cell-count count turtles
3…… if (cell-count <= 0)
4……  [ stop ]
5……end
```

2: This line counts the current created turtles (cells) and stores the this number to cell-count variable.

3 and 4: These lines stop program execution if the number of current cells are less than or equal to zero

Understanding “go” function



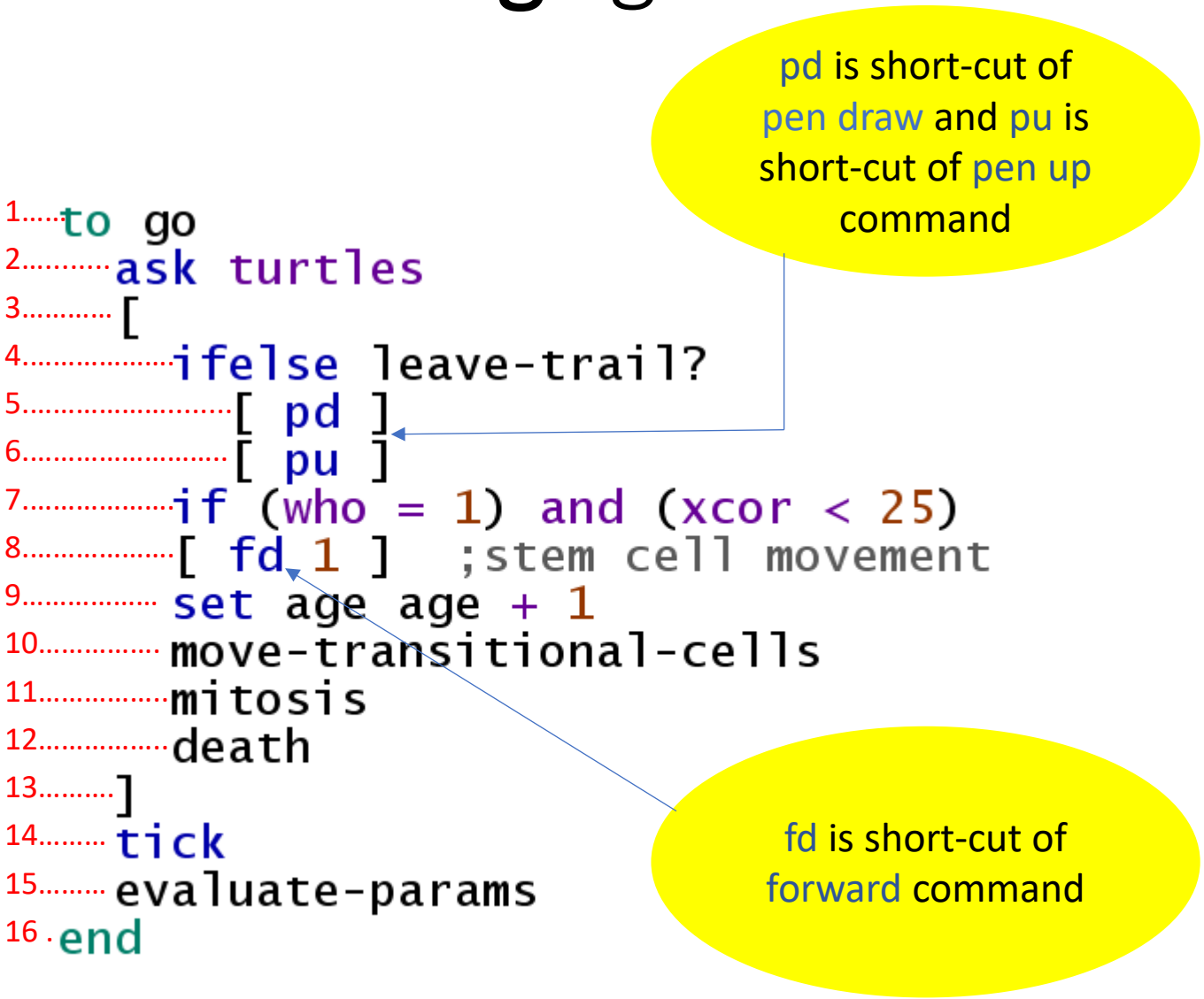
1: This line gives a name “go” to function (which is called when user clicks the go button that starts the repeatedly execution of lines 2 to 15).

2: It alerts all created turtles to be ready to execute lines 4 to 12.

4 to 6: If leave-trail switch is on, every cell draws line of turtle’s color, that shows path of the cell’s movement. If leave-trail switch is off, no cell draws line, that doesn’t show path of the cell’s movement.

7 and 8: If turtle id is 1 (i.e., the stem cell that created at first) and it is at 25 x-coordinate (regarding location) then move the turtle at one step forward.

Understanding “go” function



9: This line increase the value in age variable by 1.

10: It calls the “move-transitional-cells” function. (which moves and amplifies the transitory cells)

11: It calls the “mitosis” function. (which produces one more stem cell, moves it one step ahead, changes it’s color with various intensity of red-color, see Slide 4 for details)

12: It calls the “death” function (which kills the aged cells)

14: It increases the value of tick by

15: It calls “evaluate-params” function (which basically counts and save created number of cells and stops program execution if the number of current cells are less than or equal to zero.

THINGS TO TRY

- Try a treatment: click on the 'kill transitory cells' button while the model is running.
- This simulates treatment with a chemical agent. The agent eliminates young (red) cells that divide, and it spares older cells.
- Note that the tumor shrinks and grows again.
- Continue with this "chemotherapy" by clicking on the button again and watch the plot. Repeat the treatment several times until you have understood why it fails.

http://ccl.northwestern.edu/netlogo/models/Tumor

THINGS TO TRY…

- Most of the chemotherapy drugs known as M- and S-poisons inhibit cell division, but generally do not cause cell death.
- Thus chemotherapy can be represented as killing the young (red) transitory cells, since only young cells divide.
- Older, non-dividing transitory cells are not affected; they continue aging and finally pass away.

http://ccl.northwestern.edu/netlogo/models/Tumor

THINGS TO TRY…

- The problem is that the stem cells maintain the tumor and propagate its metastases.
- Also known as clonogenic cells, stem cells are generally resistant to chemotherapy.
- They can be eliminated only with high doses of chemicals, which endanger healthy stem cells.
- The therapeutic margin of chemical drugs is extremely narrow. That is, they do about the same amount of harm as good.

http://ccl.northwestern.edu/netlogo/models/Tumor

THINGS TO TRY…

- Let the tumor grow again.
- Click on 'kill moving stem cell' and continue running the model.
- The right blue stem cell disappears.
- Its progeny live a bit longer and then die.
- Now click on "kill original stem cell", and watch the gradual disappearance of the tumor, as no new cells are created and existing cells continue aging until they die and disappear.

http://ccl.northwestern.edu/netlogo/models/Tumor

THINGS TO TRY…

- Any questions?
- Before asking, continue exploring the model until you grasp its behavior.
- The model reveals a hidden dimension that is difficult to understand.
- It is the time dimension of tissues (and the tumor) in the body.

http://ccl.northwestern.edu/netlogo/models/Tumor

References and Sources

http://ccl.northwestern.edu/netlogo/models/Tumor



1. Make a tutorial of any model (selected from NetLogo Library) in group like the tutorial we learnt for Tumor Model (**due: 30-Nov-2021**). You will make presentations collectively. You will present your work by equally dividing slides in your group member .

- Heading size 44
- Font size: 28
- Font color black
- Background white
- Text black
- Explainer Boxes of yellow color